Heating — Cargo Compartment Heating

Cargo Compartment Low Temperature Indication After a Cold Soak

Object code	DMC-MRD-A-21-40-11-00A-420A-A	**Issue**	001-00 (new)
Issue date	2021-01-13	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Troubleshooting Engineering, Valmont (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	MERIDIAN-42 and MERIDIAN-46, all aircraft		

1. Fault description

CARGO TEMP LO displayed after a cold soak, clearing within twenty minutes of heating

The cargo compartment temperature indication reads well below the actual compartment temperature after an overnight cold soak, and normalises within about twenty minutes of the heating system running. The behaviour is repeatable on the same aircraft and absent on others in the same fleet. Where live animals are carried the indication forces the shipment to be refused at the gate, so the operational consequence is out of proportion to the technical fault. There is no correlation with compartment loading.

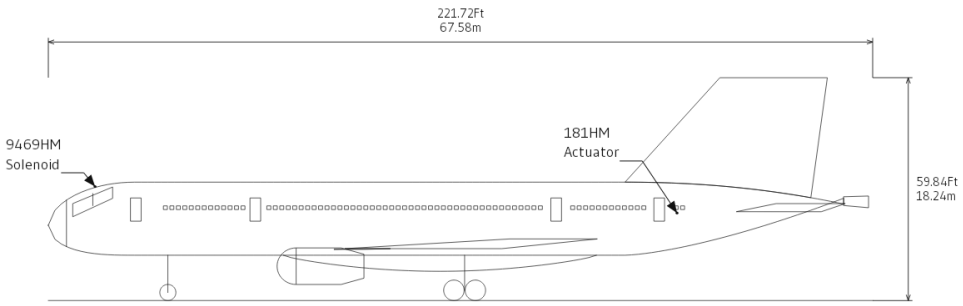
Flight deck effect: `CARGO AFT TEMP LO`, `CARGO AFT HEAT FAULT`.

Maintenance fault codes: 214011002, 214011017.

Table 1 Possible causes, most likely first

Cause	FIN	Probability
Moisture ingress at the vented transmitter body	9469HM	52 %
Water in the transmitter connector from a harness drip loop above the connector	9469HM	24 %
Heating loop actuator slow to drive from cold	181HM	15 %
Genuine loss of heating airflow to the compartment	181HM	9 %

2. Job set-up information



Heating — Cargo Compartment Heating — location, side view — Figure 1

2. Job set-up information

A. Referenced information

TYPE	DESIGNATION	IDENTIFICATION No.
GEN	The aircraft electrical power is available.	—
GEN	Access is available to zone 172, 273.	—

B. Fixtures, tools, test and support equipment

QTY	DESIGNATION	IDENTIFICATION No.
1	Portable maintenance access terminal	—

C. Consumable materials

No consumable materials required

D. Expendable parts

No expendable parts required

3. Fault isolation procedure

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STEP 1

Record the compartment temperature indication, the ambient temperature and the time since the aircraft was last powered.

ANSWER	ACTION
DONE	If this identifies the fault, the cause is: Moisture ingress at the vented transmitter body (9469HM). **Go to step 2**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 2

Measure the actual compartment temperature with a calibrated thermometer at the transmitter position.

ANSWER	ACTION
DONE	If this identifies the fault, the cause is: Water in the transmitter connector from a harness drip loop above the connector (9469HM). **Go to step 3**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 3

If the two agree, the indication is correct and the fault is in the heating system; go to step 7.

ANSWER	ACTION
DONE	If this identifies the fault, the cause is: Heating loop actuator slow to drive from cold (181HM). **Go to step 4**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 4

If the indication reads low, isolate the transmitter and measure its resistance at the connector.

ANSWER	ACTION
DONE	If this identifies the fault, the cause is: Genuine loss of heating airflow to the compartment (181HM). **Go to step 5**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

3. Fault isolation procedure

STEP 5

Compare the resistance with the temperature reference table and record both values.

ANSWER	ACTION
DONE	Continue with the next step. **Go to step 6**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 6

Inspect the transmitter body and its connector for moisture, and check whether the vent is present on the installed transmitter standard.

ANSWER	ACTION
DONE	Continue with the next step. **Go to step 7**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 7

Do the confirmation task and record the stabilisation time.

ANSWER	ACTION
DONE	Continue with the next step. **Go to step 8**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 8

Check the loop actuator 181HM drive time and the solenoid 9469HM energisation against the values in the maintenance documentation.

ANSWER	ACTION
DONE	Continue with the next step. **Go to step 9**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

4. Close-up

STEP 9

Where moisture is found at the transmitter vent or in the connector, dry the installation, re-seal the connector and repeat the confirmation task.

ANSWER	ACTION
DONE	Continue with the next step. **Go to step 10**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

STEP 10

Where the fault repeats after drying, replace the transmitter with the sealed standard and record the aircraft as modified.

ANSWER	ACTION
DONE	Continue with the next step. **Go to the end of the procedure**
NOT APPLICABLE	**Corrective action:** Go to the end of the procedure and do the confirmation task. **Go to the end of the procedure**

END OF FAULT ISOLATION PROCEDURE

Do the confirmation task AMM-M-21-40-11-700-016-A and make sure that the fault does not return.

4. Close-up

Do the confirmation task AMM-M-21-40-11-700-016-A.

Make sure that no fault message is shown and make an entry in the aircraft technical log.

Correlated fault information

Dispatch is permitted with the cargo temperature indication inoperative provided no temperature-sensitive cargo is carried. Dispatch with the compartment heating itself inoperative is subject to the route and cargo restrictions of the master minimum equipment list.

Table 2 Dispatch

MEL item	21-43-01B
Dispatchable	Yes
Condition	Cargo temperature indication inoperative, no temperature-sensitive cargo carried.